

PROMETHEUS FLAME

DRAFT REQUEST FOR SOLUTIONS



Project Title	Prometheus Flame
Project Sponsor	Department of the Air Force
Contracting Activity	Air Force Life Cycle Management Center (AFLCMC)
Questions Deadline	TBD
Response Deadline	TBD
Anticipated Project Budget	Total Phases I – II: \$20,000,000
Resultant Award Type	Prizes with potential follow-on Prototype and Production Other Transaction (OT) Agreements
Authority	10 U.S.C. § 4025, “Prizes for advanced technology achievements.” 10 U.S.C. § 4022, “Authority of the Department of Defense to Carry Out Certain Prototype Projects.”
Request for Feedback	This is a draft solicitation. We welcome your feedback! Submit recommendations NLT 12 May 2026 .

TABLE OF CONTENTS

The Problem.....	2
Our Blueprints for an Open Ecosystem	2
A True Open Competition.....	3
Challenge Phases, Prizes & Fly-Offs	3
The Mission: Your Deliverables (Early Phases)	3
Judging & Evaluation Criteria	4
Step 1: Pre-Fly-off Qualification	4
Step 2: Fly Off	5
Mission Scenario	6
Phase One: FLY OFF.....	6
Phase Two: FLY OFF	7
How to Enter	7
How do you recommend we improve this challenge?.....	8

THE PROBLEM

Open-source solutions like ArduPilot, PX4, CivTAK/ATAK, OpenCV, the Robotic Operating System (ROS), Linux, and Kubernetes have unleashed vibrant ecosystems of innovative commercial capabilities. In defense, progress remains trapped in closed, proprietary "stovepipes" where monolithic software is tightly coupled with bespoke hardware. This prevents the very companies we need—the agile, the innovative, the non-traditional—from contributing to national security.

To kindle an open-source community and ignite innovation at scale, the Mission Systems ASE Group is launching **Prometheus Flame**.

The character of warfare is being reshaped by software. To meet this change, the Department of War (DoW) must revolutionize how it develops and deploys software for Autonomy and Mission Systems applications.

The **Prometheus Flame Challenge** seeks to burn this vendor locked paradigm to the ground and shed light on the potential of open-source enthusiasts and non-traditional vendor ecosystems to develop and field standards-compliant dual-use Autonomy and Mission Systems product lines. This will create the "flywheel" effect to bootstrap robust commercial and military technologies, lowering cost and increasing capabilities for U.S. and its allies. This initiative directly supports the strategic imperatives to create mass, get humans out of harm's way, accelerate innovations for commercial applications where military dual use exists, and burn down the cost curve by leveraging dual-use Autonomy and Mission Systems capabilities for scalable, affordable effects.

*"Our goal for **Prometheus Flame** is to create an open-source, enthusiast/dual-use-grade learning platform that establishes a new, vibrant community around standards-compliant Autonomy and Mission Systems product lines."*

OUR BLUEPRINTS FOR AN OPEN ECOSYSTEM

To that end, the DoW is publishing two "Government Reference Architecture" (GRA) standards that form the foundation for open, modular, agile, and hardware-agnostic capabilities for next-generation dual-use technologies:

Agile Mission Suite GRA (AMS GRA): Defines a software-defined, modular open architecture for mission system capabilities (e.g., sensors, communication, data fusion, etc.).

Autonomy GRA (A-GRA): Defines standardized behaviors, decision-making, and teaming for autonomous systems.

These standards are currently Controlled Unclassified Information (CUI); however, the Government is in the process of downgrading them to Unclassified / Public Release. When publicly released, they will be posted to <https://github.com/modular-af> on or around 1 June 2026. In the interim, Offerors may request the CUI version of the standards by sending an email (see Contact Information section) with documentation that shows they are authorized to handle CUI materials. The Government team will then coordinate the secure delivery of the requested standards. Alternatively, Offerors may download the publicly available [Open Mission Systems \(OMS\)](#) and [Universal C2 Interface \(UCI\)](#) standards for reference as both AMS GRA and A-GRA's foundation are built upon those standards.

A TRUE OPEN COMPETITION

This is not a typical solicitation. **Prometheus Flame** is a flexible, multi-phase competition designed to identify and fund the best partners to build the foundation of a vibrant new dual-use technology ecosystem.

This is an open competition. **Offerors may enter at any time.** Participation in a prior phase is not required. Whether you are starting with a fresh idea or have an existing prototype, there is an on-ramp for you. At each phase, the goal is to submit as complete a version of the final product as possible for that stage of development.

CHALLENGE PHASES, PRIZES & FLY-OFFS

The **Prometheus Flame** is composed of a continuous challenge effort composed of 3-6 month long phases. Each phase will culminate in a “fly-off” demonstration of each solution in Dayton, Ohio, with the top teams earning a prize.

This RFS will be updated with specific dates once the AMS GRA and A-GRA standards are publicly released. That will start the clock (i.e., T-0) for this challenge.

The first phases of this challenge will focus on building a foundation of open-source software implementation for both AMS GRA and A-GRA. Once that foundation is firmly established, subsequent phases will leverage that to identify best-of-breed Mission Systems and Autonomy skills such as weather management, cooperative search and rescue, etc.

Phase	Date	Prize Total (\$)	Num Prizes ¹	OT Total (\$)
1	~T+3 months	\$6,000,000	Up to 10	\$4,000,000
2	~T+6 months	\$6,000,000	Up to 10	\$4,000,000
...	...	...	...	...
N	~T+n*(3 months)	TBD	TBD	...

Our commitment does not end with the prizes. The **Prometheus Flame** is a valuable mechanism to identify long-term strategic partners. We intend to award Other Transaction Agreements to teams whose solutions are the most compelling and strategically aligned with our vision and long-term needs. This will provide a recurring funding stream to mature the open-source foundation, lead its proliferation, and support the rapidly growing AMS GRA and A-GRA ecosystem. All participants, regardless of the phase they enter, will be evaluated for this opportunity.

¹To qualify for prizes, a solution must prove AMS GRA and/or A-GRA compliance and be delivered with a permissive open-source license (e.g., MIT, Apache, BSD, Unlicense, etc.) for public release. Fewer prizes and OTAs may be awarded if submitted solutions do not meet these criteria.

THE MISSION: YOUR DELIVERABLES (EARLY PHASES)

Our goal for the initial phases of the **Prometheus Flame** is to create an open-source, enthusiast/dual-use-grade learning platform foundation that establishes a vibrant, self-sustaining community—a “flywheel effect”—around standards-compliant Autonomy and Mission Systems product lines. This continuous prize challenge will serve as the primary mechanism to identify and fund the best partners to build the foundation

of this new defense ecosystem, enabling hobbyists, academia, non-traditional, and traditional vendors to build dual-use capabilities.

Competitive solutions should include the following public-release content:

- **Open-Source Implementations:** Open-source AMS GRA and A-GRA-compliant implementations that can be simulated and flown on a variety of commercial-off-the-shelf (COTS) hardware solutions. *(Note: This challenge is focused on software development and documentation. It is not intended to mature hardware solutions but rather should leverage existing hardware solutions available on the open market.)*
- **Turn-Key Projects:** End-to-end example AMS GRA and A-GRA-compliant projects (e.g., "hello world," communications, weather management, cooperative autonomous search, etc.).
- **Public-Facing Resource:** A source code repository, public website, getting-started guides, comprehensive documentation, and instructional videos.

Ultimately, the most competitive solutions in the initial phases of this challenge will be those that best empower third parties to readily leverage the provided tools to create their own new, innovative, and standard-compliant products.

JUDGING & EVALUATION CRITERIA

Judging will heavily favor solutions that are not only technically proficient but are also easy to use, well-documented, and extensible. Each phase will culminate in a fly-off event where solutions are evaluated across critical areas by independent parties. Results will be posted to a public forum.

Step 1 and 2 Timeline	
RFS	~T+0 months
Step 1	~T+1.5 months
Step 2	~T+3 months

The evaluation will be completed in two steps: (1) Pre-Fly-Off Qualification & (2) Fly Off

STEP 1: PRE-FLY-OFF QUALIFICATION

This initial qualification step requires the submission of the following materials NLT two months after the Government posts the final solicitation:

- 5-page / 12-point-font white paper
- AMS GRA and A-GRA compliance report for each Minimum Procurable Units (MPUs) or architectural element developed for the Fly Off
- Video demonstrating your code running in simulation and/or in COTS flight-test hardware
- Hardware and Software Bill of Materials
- Access to current code baseline

The Government will evaluate the above materials against the criteria outlined in the "Evaluation Factors" section to select up to 20 submissions for the Fly Off.

EVALUATION FACTORS

The Government will evaluate Solutions in accordance with the established criteria. Based on the evaluation, the Government will categorize Solutions as “Selectable Plus,” “Selectable,” or “Not Selectable.”

- **Selectable Plus:** Vendor has adequate approach regarding their solution to solve the objectives of this RFS and is best suited to meet the needs of the Government requirement.
- **Selectable:** Vendor has adequate approach regarding their solution to solve the objectives of this RFS.
- **Not Selectable:** Vendor does not have adequate approach regarding their solution to solve the objectives of this RFS.

The Government will notify unsuccessful offerors, either via email or letter, to inform them that the proposed effort is not selected at this time.

Category	Description
1. Technical Solution	The offeror provides adequate technical approach to include viable CI/CD pipelines.
2. Risk	The offeror identifies risk and proposes viable mitigation plans.

STEP 2: FLY OFF

Offerors must provide the following data NLT 24 hours prior to Fly Off:

- All software source code, CI/CD pipelines, and documentation
- AMS GRA and A-GRA compliance report for each Minimum Procurable Units (MPUs) or architectural element developed for the Fly Off
- 5 page / 12-point-font Work Plan for first Project Order of follow-on work (include 3-month base effort, timeline, specific deliverables, and milestone payments)
- Price proposal for the Work Plan (Firm Fixed Price)

The Government will evaluate the above materials against the criteria outlined in the “Evaluation Factors” section to identify Prize and OT awards.

EVALUATION FACTORS

The Government will evaluate Solutions in accordance with the established criteria. Based on the evaluation, the Government will categorize Solutions as “Exceeds,” “Meets,” or “Does Not Meet.”

- **Exceeds:** Offeror has excellent approach regarding their solution to solve the objectives and is best suited to meet the mission requirement. Highly recommended and prioritized for Fly Off or OT award, subject to available funds.
- **Meets:** Offeror has adequate approach regarding their solution to solve the objectives.
- **Does Not Meet:** Offeror does not have adequate approach regarding their solution to solve the objectives of this RFS.

The Government will notify unsuccessful offerors, either via email or letter, to inform them that the proposed effort is not selected at this time.

Category	Description
1. Mission Performance	The Government will evaluate the extent to which the software demonstrates successful mission execution through simulated and/or live-flight demonstrations while providing a compelling, reliable technical solution with unique military or dual-use utility.
2. Usability & Onboarding	The Government will evaluate the extent to which the software architecture, APIs, and supporting documentation facilitate accessible third-party modification and integration, supported by diverse, reliable, and turnkey example projects that effectively demonstrate core capabilities.
3. Technical Implementation & Standards Compliance	The Government will evaluate the extent to which the software demonstrates clean, modular code and architectural compliance with AMS GRA/A-GRA standards through comprehensive documentation and a reliable simulation environment for verification.
4. Risk	The Government will evaluate the extent to which the vendor identifies risk and proposes viable mitigation plans.

MISSION SCENARIO

The first two Phases of Prometheus Flame will be a multi-stage gauntlet designed to prove that your open-source AMS and A-GRA implementations exist as true, functional software rather than smoke and mirrors. The goal is to demonstrate absolute portability and modularity, proving that your code can seamlessly drive a platform without the burden of vendor lock-in.

PHASE ONE: FLY OFF

SIMULATION

Participants will deploy their open-source stack into a Government-owned virtual arena. We are looking for a demonstration of AMS compliance, meaning your software system, skills, and services must meet MPU and documentation requirements. During the simulation, your A-GRA and AMS Skills will be put to the test. A generic Mission System will issue standard commands, and your autonomy skills must translate those into precise flight primitives. Success here proves your architecture is sound and ready for the real world.

MODULARITY TEST

The true climax of the fly-off is the live architecture audit and the “hot-swap” test. Evaluators will inspect the code to ensure no platform-specific shams or proprietary libraries were used to bypass the open-source GRA standards. To definitively prove modularity, the Government team may command a mid-mission swap. You

might be asked to swap out a sensor-processing service or an autonomy skill to prove that the system can be upgraded or altered on demand. Winning the fly-off means your code proved that the USAF can finally swap solutions and upgrade capabilities without breaking the underlying mission system.

PHASE TWO: FLY OFF

Participants will be handed a physical, Government-furnished drone—or a standardized hardware kit—and given a strict integration window. This is where the portability of your code is proven. You must port your open-source GRA stack directly onto the vehicle's flight controller and compute payload. During the live flight, the drone must autonomously execute mission-impactful tasks. The evaluators will be watching to ensure the autonomy skills (A-GRA) and system services (AMS) are communicating perfectly in real-time, dictating the vehicle's behavior without error.

HOW TO ENTER

Simply send an email (see Contact Information section) with the following information:

- Offeror Name
- Proof of United States or US ally-based headquarters and ownership. This Competition is open to industry and academia.
- Official GitHub account names

The Government will create an Offeror-specific private Git repository where you will post your solutions. Entrance into competition does not guarantee a position at the Fly Off in Dayton, OH.

CONTACT INFORMATION

Email Address: aflcmc.open-source.challenge@us.af.mil

Primary Contracting Officer: Mr. Joseph Mizzi (AFLCMC)

Program Manager: 1st Lt Justin Brogden (AFLCMC)

HOW DO YOU RECOMMEND WE IMPROVE THIS CHALLENGE?

This RFS is still in the draft form. We welcome feedback! Link: <https://forms.osi.apps.mil/r/J4LKTevZ29>

We intend to release the final version of this solicitation on or around 1 June 2026 after the formal public Unclassified release of AMS GRA and A-GRA. That enables multiple rounds of solicitation drafts in the interim for Offerors' feedback. For this initial draft, **we respectfully request feedback (see Contact Information) no later than 12 May 2026**. We intend to update the draft and provide an additional opportunity for industry to provide feedback before the final solicitation is released. NOTE: Providing responses to this RFI is completely VOLUNTARY. The Government will NOT reimburse any costs associated with providing a response.



FOLLOW ON OTHER TRANSACTION 10 USC 4022

Initial follow on work will be building the knowledge base and ecosystem for AMS and A-GRA.

Work will include but not be limited to:

- Starter kit development including reference implementations, documentation, and outreach
- Automated CI/CD and compliance pipelines
- Cyber patching
- Architectural consortia
- Support current and future adoption
- Implement operational capability